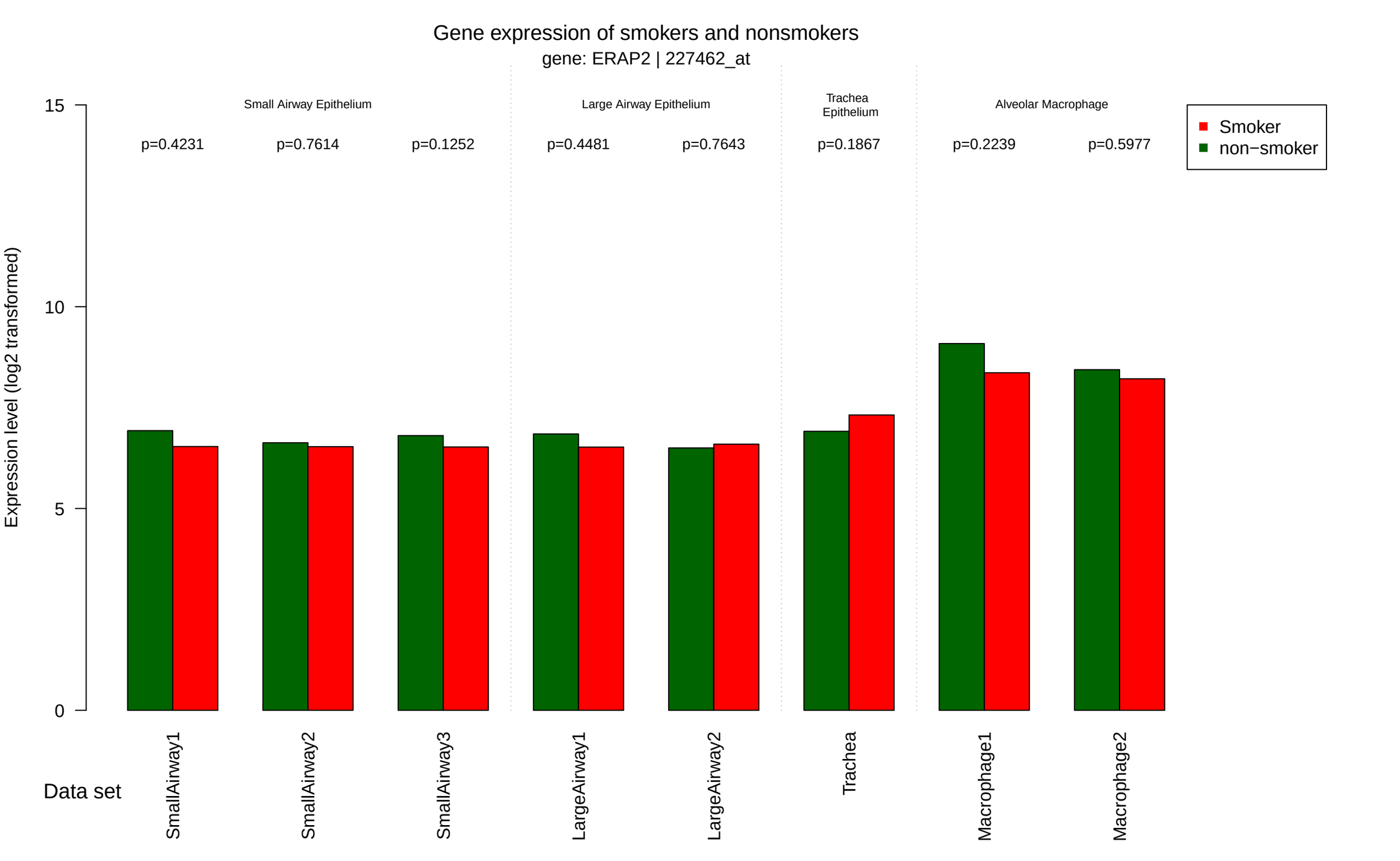
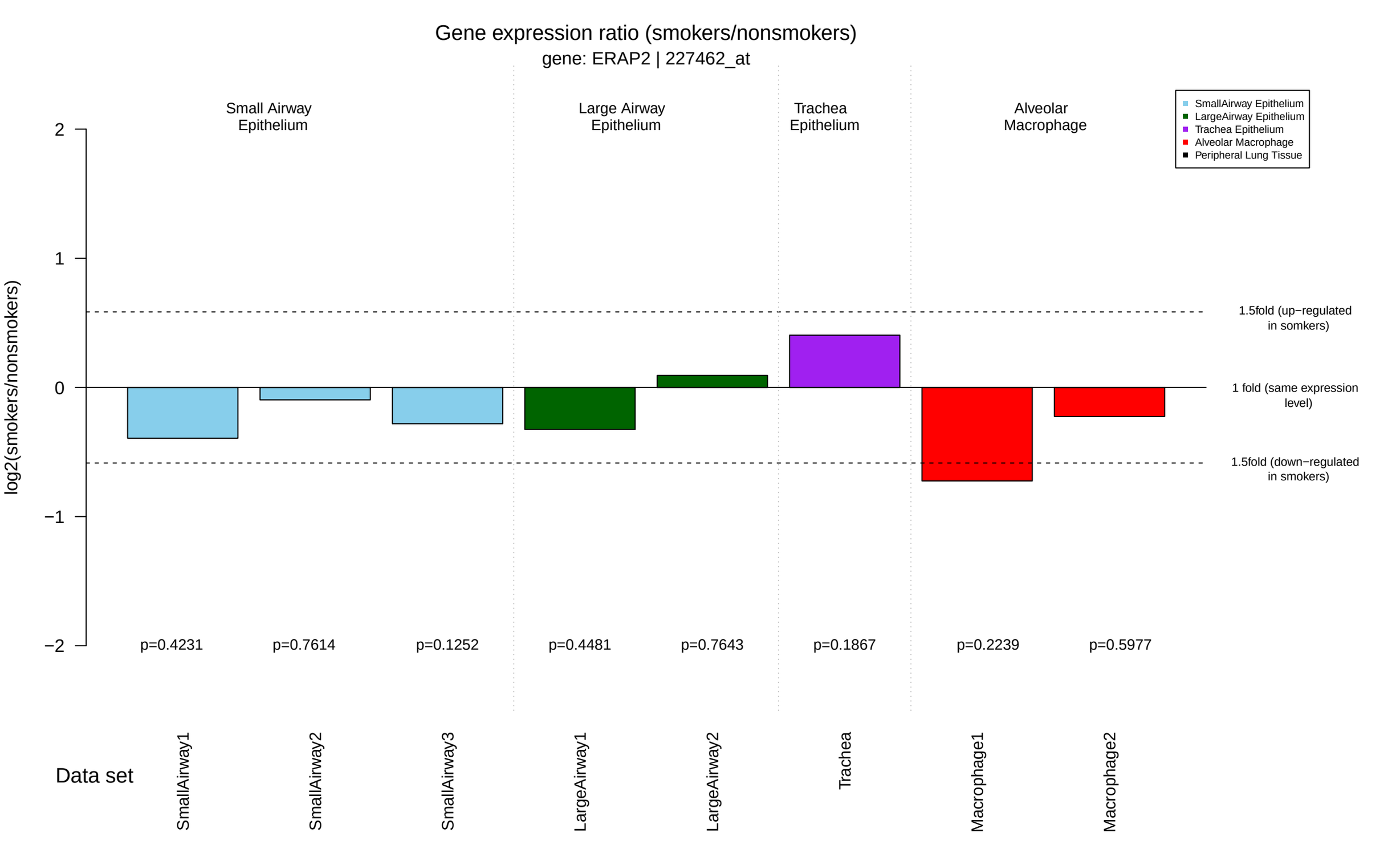
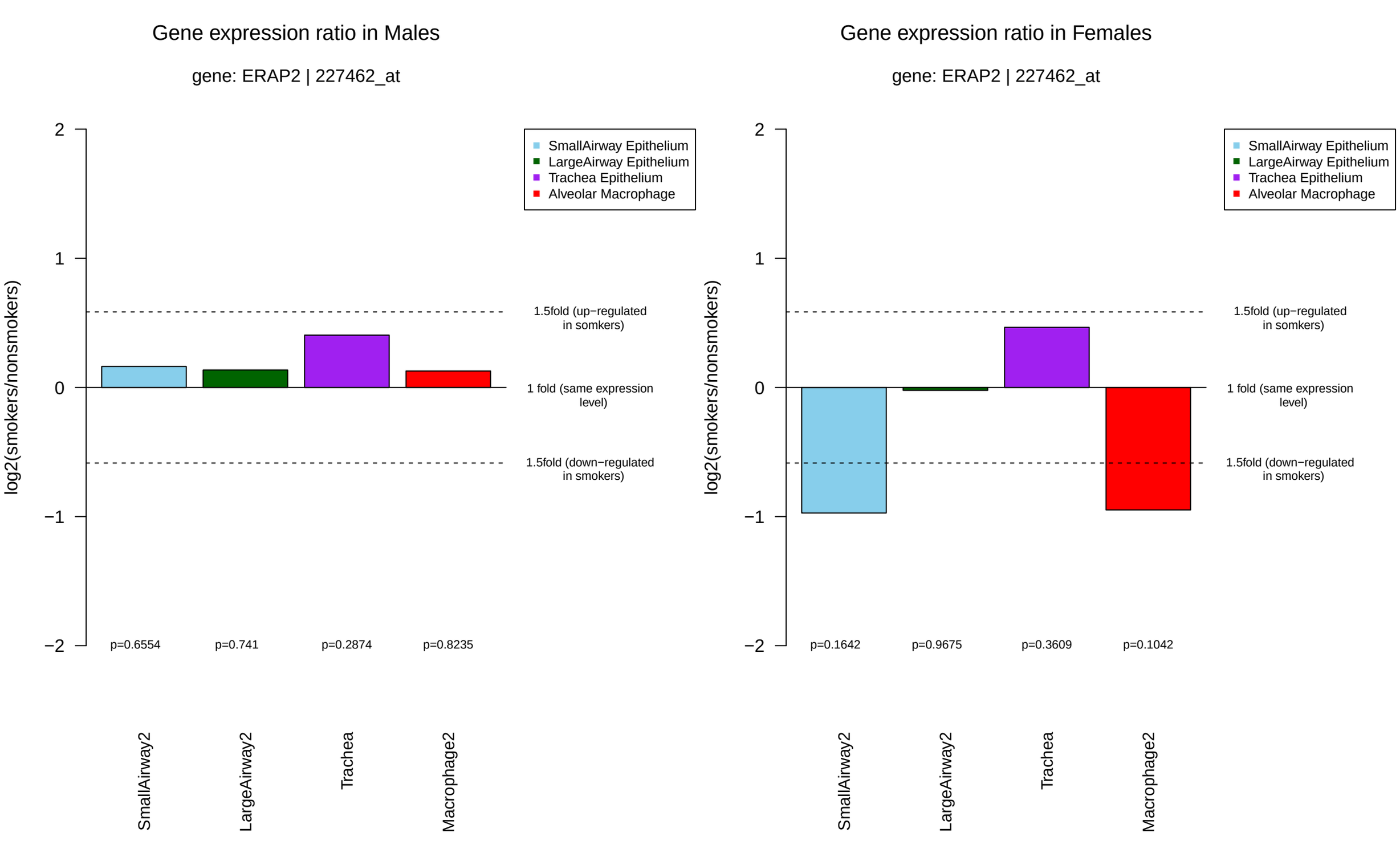
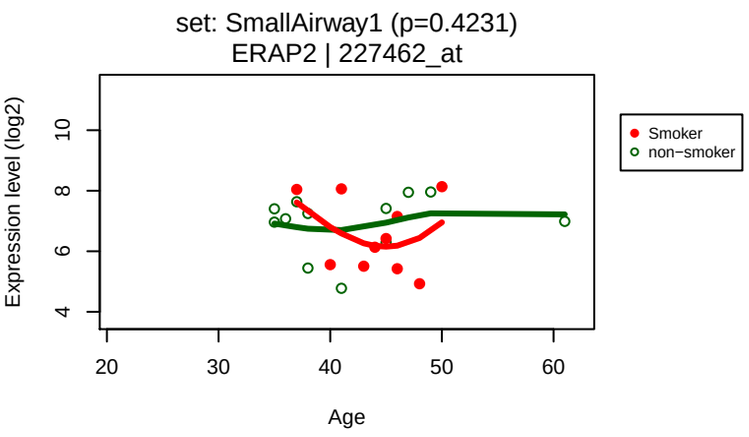
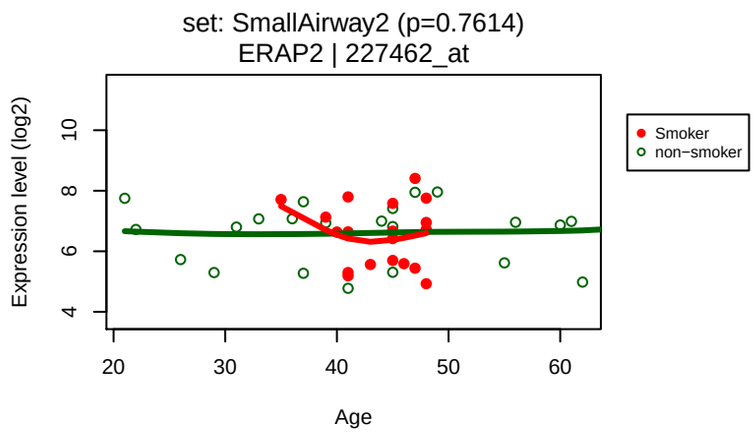
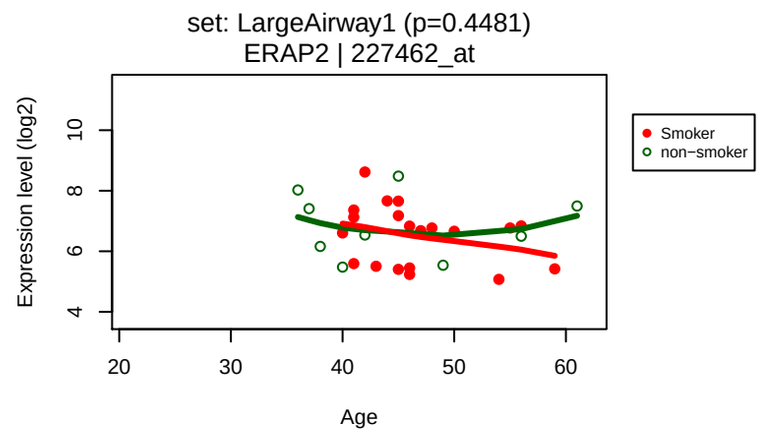

Gene expression of smokers and nonsmokers
gene: ERAP2 | 227462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4231
p=0.7614
p=0.1252
p=0.4481
p=0.7643
p=0.1867
p=0.2239
p=0.5977
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

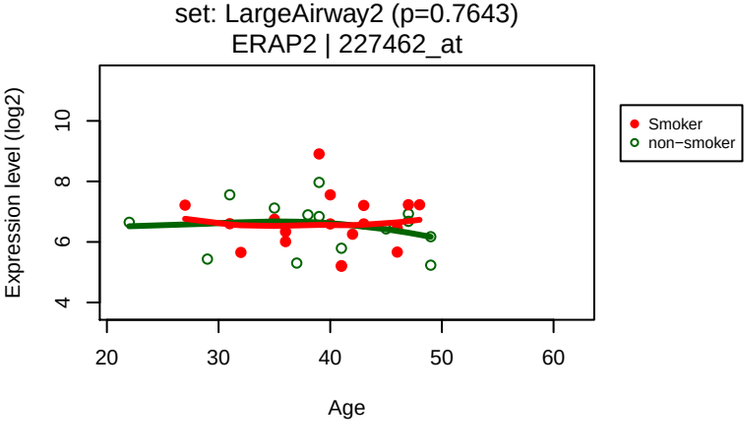

Gene expression ratio (smokers/nonsmokers)
gene: ERAP2 | 227462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4231
p=0.7614
p=0.1252
p=0.4481
p=0.7643
p=0.1867
p=0.2239
p=0.5977
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

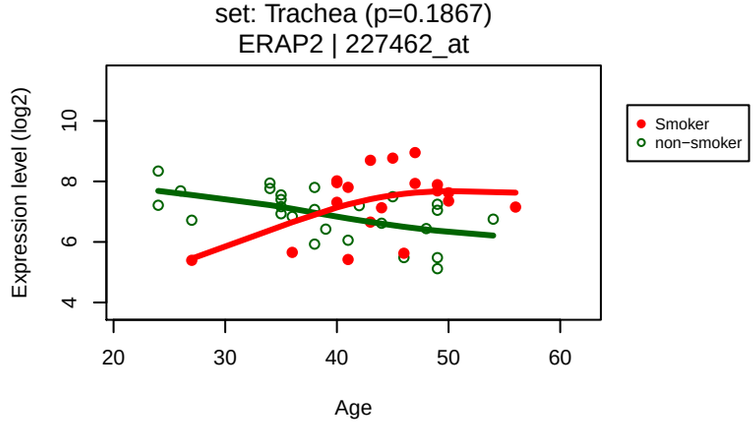

Gene expression ratio in Males
gene: ERAP2 | 227462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6554
p=0.741
p=0.2874
p=0.8235
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP2 | 227462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1642
p=0.9675
p=0.3609
p=0.1042
SmallAirway2
LargeAirway2
Trachea
Macrophage2

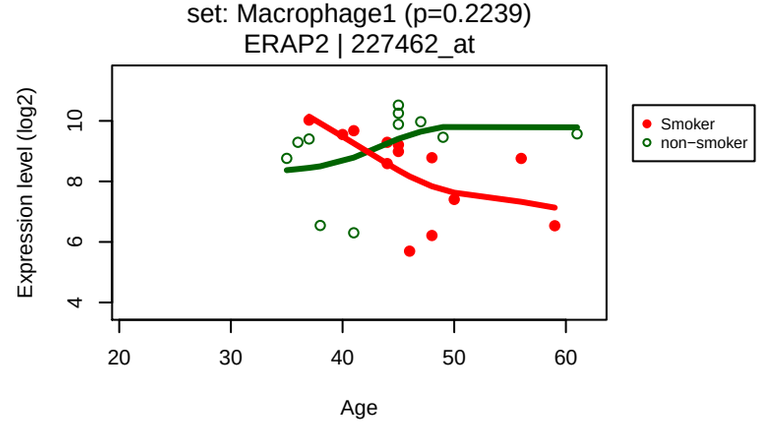

set: SmallAirway1 (p=0.4231)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker

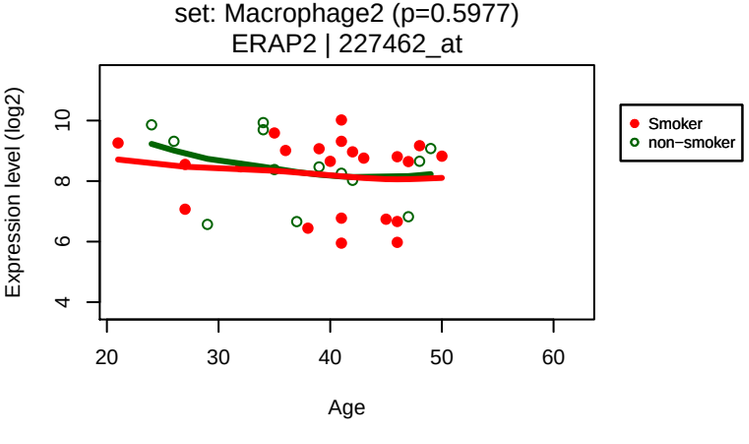

set: SmallAirway2 (p=0.7614)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker

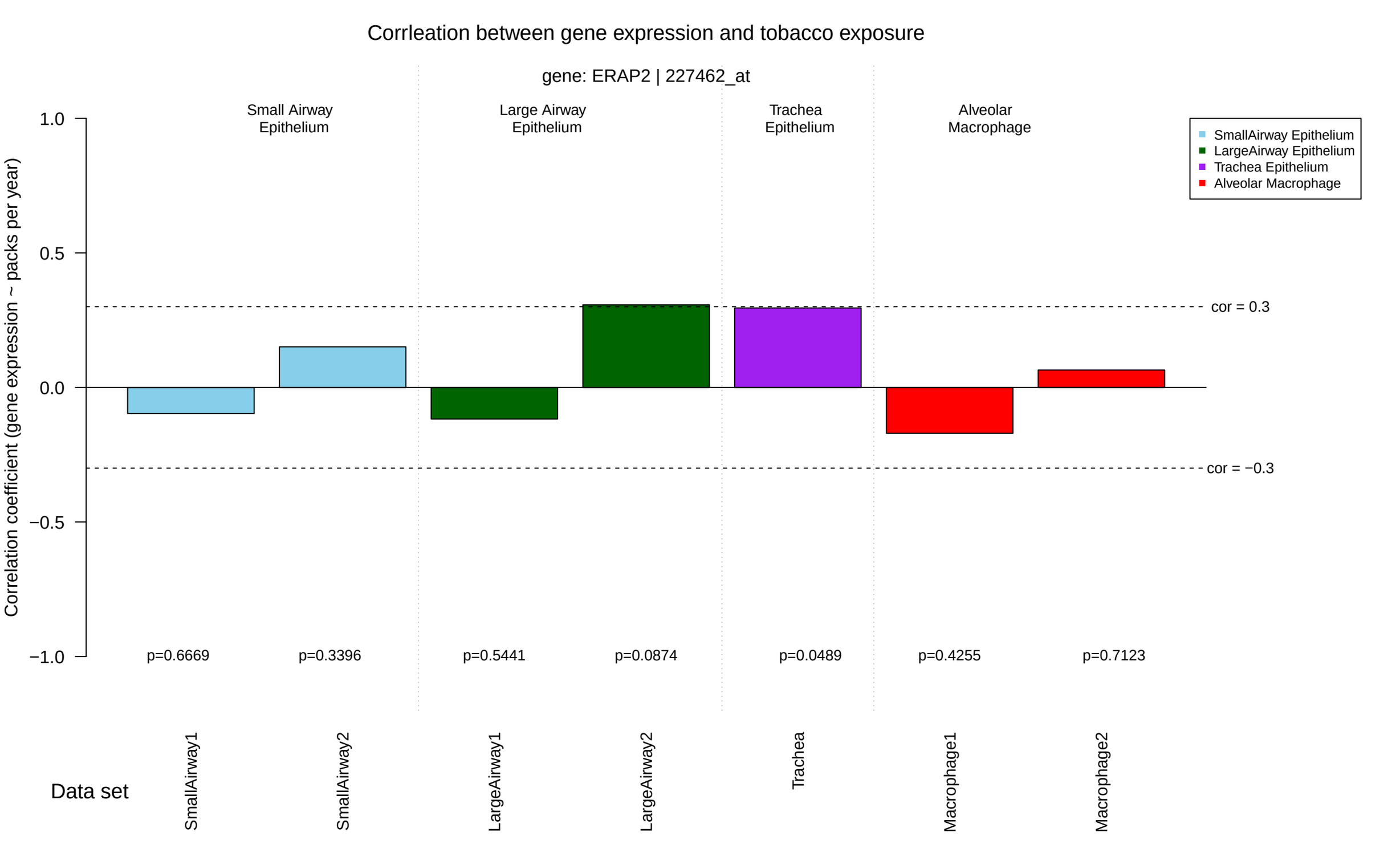

set: LargeAirway1 (p=0.4481)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7643)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1867)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2239)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5977)
ERAP2 | 227462_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP2 | 227462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6669
p=0.3396
p=0.5441
p=0.0874
p=0.0489
p=0.4255
p=0.7123
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2